Trigonometric Functions and Circular Measure

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Trigonometric Functions & Circular Measure Practice Exercises**

The diagram shows a sector of a circle with centre O. The arc length of the sector is given by 6π cm and the area of the sector is 27π cm^2. The radius of the circle is r cm and the angle of the sector is θ radians. Find the value of r and θ.

Final Answer: r = 9, $\theta = \frac{2\pi}{3}$. The formula for the arc length of a sector is $L = r\theta$, where L is the arc length, r is the radius, and θ is the angle in radians. We are given that $L = 6\pi$. So, we have the equation: $r\theta = 6\pi$ (1) The formula for the area of a sector is $A = \frac{1}{2}r^2\theta$, where A is the area, r is the radius, and θ is the angle in radians. We are given that $A = 27\pi$. So, we have the equation: $\frac{1}{2}r^2\theta = 27\pi$ (2) We can r.

2 **[Foundational]** • **Target: Radian Measure, Arc Length and Sectors**

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is 6 cm and the radius is 3 cm, find the angle θ in radians.

Final Answer: 2 radians. The formula relating arc length (s), radius (r), and angle (θ) in radians is $s = r\theta$. We are given $s = 6$ cm and $r = 3$ cm. Substituting these values into the formula, we get $6 = 3\theta$. To find θ, we divide both sides by 3: $\theta = \frac{6}{3} = 2$ radians. Therefore, the angle at the centre is 2 radians.

3 **[Foundational]** • **Target: Trigonometric Identities and Ratios**

A sector has radius 11 cm and angle 1.1 rad. Find the arc length.

Final Answer: 121/10. Applying Circular Measure principles, $s = r\theta = 11 \times 1.1 = 121/10$ cm Therefore, the verified correct option is 121/10.

4 **[Foundational]** • **Target: Solving Trigonometric Equations**

A sector has radius 13 cm and angle 2.3 rad. Find the arc length.

Final Answer: 299/10. Applying Circular Measure principles, $s = r\theta = 13 \times 2.3 = 299/10$ cm Therefore, the verified correct option is 299/10.

5 **[Foundational]** • **Target: Double Angle Formulae and Graph Amplitude**

A sector has radius 13 cm and angle 1.7 rad. Find the arc length.

Final Answer: 221/10. Applying Circular Measure principles, $s = r\theta = 13 \times 1.7 = 221/10$ cm Therefore, the verified correct option is 221/10.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A sector has radius 19 cm and angle 0.9 rad. Find the arc length.

Final Answer: 171/10. Applying Circular Measure principles, $s = r\theta = 19 \times 0.9 = 171/10$ cm Therefore, the verified correct option is 171/10.

7 [Foundational] • Target: Radian Measure, Arc Length and Sectors

A sector has radius 11 cm and angle 0.3 rad. Find the arc length.

Final Answer: 33/10. Applying Circular Measure principles, $s = r\theta = 11 \times 0.3 = 33/10$ cm Therefore, the verified correct option is 33/10.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Trigonometric Identities and Ratios

A sector has radius 14 cm and angle 1.9 rad. Find the arc length.

Final Answer: 133/5. Applying Circular Measure principles, $s = r\theta = 14 \times 1.9 = 133/5$ cm Therefore, the verified correct option is 133/5.

9 [Intermediate] • Target: Solving Trigonometric Equations

A sector has radius 26 cm and angle 3.1 rad. Find the arc length.

Final Answer: 403/5. Applying Circular Measure principles, $s = r\theta = 26 \times 3.1 = 403/5$ cm Therefore, the verified correct option is 403/5.

10 [Intermediate] • Target: Double Angle Formulae and Graph Amplitude

A sector has radius 14 cm and angle 1.6 rad. Find the arc length.

Final Answer: 112/5. Applying Circular Measure principles, $s = r\theta = 14 \times 1.6 = 112/5$ cm Therefore, the verified correct option is 112/5.

11 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A sector has radius 26 cm and angle 1.4 rad. Find the arc length.

Final Answer: 182/5. Applying Circular Measure principles, $s = r\theta = 26 \times 1.4 = 182/5$ cm Therefore, the verified correct option is 182/5.

12 [Intermediate] • Target: Radian Measure, Arc Length and Sectors

A sector has radius 9 cm and angle 0.3 rad. Find the arc length.

Final Answer: 27/10. Applying Circular Measure principles, $s = r\theta = 9 \times 0.3 = 27/10$ cm Therefore, the verified correct option is 27/10.

© ClipSAT — clipsat.org | All Rights Reserved

13 [Intermediate] • Target: Trigonometric Identities and Ratios

A sector has radius 16 cm and angle 1.1 rad. Find the arc length.

Final Answer: 88/5. Applying Circular Measure principles, $s = r\theta = 16 \times 1.1 = 88/5$ cm Therefore, the verified correct option is 88/5.

14 [Intermediate] • Target: Solving Trigonometric Equations

A sector has radius 7 cm and angle 0.4 rad. Find the arc length.

Final Answer: 14/5. Applying Circular Measure principles, $s = r\theta = 7 \times 0.4 = 14/5$ cm Therefore, the verified correct option is 14/5.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Double Angle Formulae and Graph Amplitude • Cross-Lesson: 6.01 Permutations and Combinations

A sector has radius 8 cm and angle 1.8 rad. Find the arc length.

Final Answer: 72/5. Applying Circular Measure principles, $s = r\theta = 8 \times 1.8 = 72/5$ cm Therefore, the verified correct option is 72/5.

16 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

A sector has radius 6 cm and angle 1.4 rad. Find the arc length.

Final Answer: 42/5. Applying Circular Measure principles, $s = r\theta = 6 \times 1.4 = 42/5$ cm Therefore, the verified correct option is 42/5.

17 [Challenging] • Target: Radian Measure, Arc Length and Sectors • Cross-Lesson: 8.01 Integration and Kinematics

A sector has radius 6 cm and angle 2.6 rad. Find the arc length.

Final Answer: 78/5. Applying Circular Measure principles, $s = r\theta = 6 \times 2.6 = 78/5$ cm Therefore, the verified correct option is 78/5.

18 [Challenging] • Target: Trigonometric Identities and Ratios • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

A sector has radius 7 cm and angle 2.6 rad. Find the arc length.

Final Answer: 91/5. Applying Circular Measure principles, $s = r\theta = 7 \times 2.6 = 91/5$ cm Therefore, the verified correct option is 91/5.

© ClipSAT — clipsat.org | All Rights Reserved

19 [Challenging] • Target: Solving Trigonometric Equations • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

A sector has radius 9 cm and angle 2.5 rad. Find the arc length.

Final Answer: 45/2. Applying Circular Measure principles, $s = r\theta = 9 \times 2.5 = 45/2$ cm Therefore, the verified correct option is 45/2.

20 [Challenging] • Target: Double Angle Formulae and Graph Amplitude • Cross-Lesson: 3.01 Indices, Surds and Polynomials

A sector has radius 6 cm and angle 0.3 rad. Find the arc length.

Final Answer: 9/5. Applying Circular Measure principles, $s = r\theta = 6 \times 0.3 = 9/5$ cm Therefore, the verified correct option is 9/5.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius $r = 6 \text{ cm}$ and central angle $\theta = \frac{\pi}{3} \text{ rad}$. What is the arc length of the sector?

a. $\frac{6\pi}{3} \text{ cm}$ ✓

b. $\frac{36\pi}{6} \text{ cm}$

c. $6\pi \text{ cm}$

d. $\frac{6\pi}{6} \text{ cm}$

Arc length formula in radians: $s = r\theta = (6)\left(\frac{\pi}{3}\right) = \frac{6\pi}{3} \text{ cm}$.

22 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius $r = 8 \text{ cm}$ and central angle $\theta = \frac{\pi}{4} \text{ rad}$. What is the arc length of the sector?

a. $\frac{8\pi}{4} \text{ cm}$ ✓

b. $\frac{64\pi}{8} \text{ cm}$

c. $8\pi \text{ cm}$

d. $\frac{8\pi}{8} \text{ cm}$

Arc length formula in radians: $s = r\theta = (8)\left(\frac{\pi}{4}\right) = \frac{8\pi}{4} \text{ cm}$.

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius $r = 10 \text{ cm}$ and central angle $\theta = \frac{\pi}{5} \text{ rad}$. What is the arc length of the sector?

a. **$\frac{10\pi}{5} \text{ cm}$** ✓

b. $\frac{100\pi}{10} \text{ cm}$

c. $10\pi \text{ cm}$

d. $\frac{10\pi}{10} \text{ cm}$

Arc length formula in radians: $s = r\theta = (10)\left(\frac{\pi}{5}\right) = \frac{10\pi}{5} \text{ cm}$.

24 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius $r = 12 \text{ cm}$ and central angle $\theta = \frac{\pi}{6} \text{ rad}$. What is the arc length of the sector?

a. **$\frac{12\pi}{6} \text{ cm}$** ✓

b. $\frac{144\pi}{12} \text{ cm}$

c. $12\pi \text{ cm}$

d. $\frac{12\pi}{12} \text{ cm}$

Arc length formula in radians: $s = r\theta = (12)\left(\frac{\pi}{6}\right) = \frac{12\pi}{6} \text{ cm}$.

25 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius $r = 14 \text{ cm}$ and central angle $\theta = \frac{\pi}{7} \text{ rad}$. What is the arc length of the sector?

a. $\frac{196\pi}{14} \text{ cm}$

b. **$\frac{14\pi}{7} \text{ cm}$** ✓

c. $14\pi \text{ cm}$

d. $\frac{14\pi}{14} \text{ cm}$

Arc length formula in radians: $s = r\theta = (14)\left(\frac{\pi}{7}\right) = \frac{14\pi}{7} \text{ cm}$.

26 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius $r = 16 \text{ cm}$ and central angle $\theta = \frac{\pi}{8} \text{ rad}$. What is the arc length of the sector?

a. **$\frac{16\pi}{8} \text{ cm}$** ✓

b. $\frac{256\pi}{16} \text{ cm}$

c. $16\pi \text{ cm}$

d. $\frac{16\pi}{16} \text{ cm}$

Arc length formula in radians: $s = r\theta = (16)\left(\frac{\pi}{8}\right) = \frac{16\pi}{8} \text{ cm}$.

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius \(r = 18 \text{ cm} \) and central angle \(\theta = \frac{\pi}{9} \text{ rad} \). What is the arc length of the sector?

a. **\(\frac{18\pi}{9} \text{ cm} \)** ✓

b. \(\frac{324\pi}{18} \text{ cm} \)

c. \(18\pi \text{ cm} \)

d. \(\frac{18\pi}{18} \text{ cm} \)

Arc length formula in radians: \(s = r\theta = (18)\left(\frac{\pi}{9}\right) = \frac{18\pi}{9} \text{ cm} \).

28 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius \(r = 20 \text{ cm} \) and central angle \(\theta = \frac{\pi}{10} \text{ rad} \). What is the arc length of the sector?

a. \(20\pi \text{ cm} \)

b. \(\frac{400\pi}{20} \text{ cm} \)

c. **\(\frac{20\pi}{10} \text{ cm} \)** ✓

d. \(\frac{20\pi}{20} \text{ cm} \)

Arc length formula in radians: \(s = r\theta = (20)\left(\frac{\pi}{10}\right) = \frac{20\pi}{10} \text{ cm} \).

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Radian Measure, Arc Length and Sectors

A circle of radius 8 cm has a central sector with angle $\theta = \frac{3\pi}{4}$ rad. Find the exact arc length of the sector.

a. 6π cm ✓

b. 12π cm

c. 4π cm

d. 24π cm

Arc length $s = r\theta = 8\left(\frac{3\pi}{4}\right) = 6\pi$ cm.

30 [Intermediate] • Target: Radian Measure, Arc Length and Sectors

Find the area of a sector of radius 6 cm having perimeter 22 cm.

a. 30 cm^2 ✓

b. 36 cm^2

c. 24 cm^2

d. 18 cm^2

Perimeter $= 2r + s = 12 + s = 22 \Rightarrow s = 10$ cm. Area $= \frac{1}{2}rs = \frac{1}{2}(6)(10) = 30$ cm^2.

© ClipSAT — clipsat.org | All Rights Reserved

31 [Intermediate] • Target: Radian Measure, Arc Length and Sectors

A chord of length 10 cm subtends an angle of 2θ at the center of a circle of radius 13 cm. Find $\sin\theta$.

a. $\frac{5}{13}$ ✓

b. $\frac{12}{13}$

c. $\frac{10}{13}$

d. $\frac{5}{12}$

The chord is bisected: half-chord $= 5$ cm. In the right triangle, $\sin\theta = \frac{5}{13}$.

32 [Intermediate] • Target: Trigonometric Identities and Ratios

Simplify the trigonometric expression $\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}$.

a. $\sec\theta\csc\theta$ ✓

b. $\tan\theta$

c. 1

d. $\sec^2\theta$

$\frac{\sin^2\theta+\cos^2\theta}{\sin\theta\cos\theta} = \frac{1}{\sin\theta\cos\theta} = \csc\theta\sec\theta$.

33 [Intermediate] • Target: Trigonometric Identities and Ratios

Given that $\tan\theta = -\frac{3}{4}$ and θ lies in the second quadrant, find the value of $\cos\theta$.

a. $-\frac{4}{5}$ ✓

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $\frac{3}{5}$

In quadrant 2, cosine is negative. With opposite 3 and adjacent 4, hypotenuse is 5: $\cos\theta = -4/5$.

34 [Intermediate] • Target: Trigonometric Identities and Ratios

Prove the identity $\sec^2\theta - \tan^2\theta = 1$. Which of the following equals $(\sec\theta - \tan\theta)(\sec\theta + \tan\theta)$?

a. 1 ✓

b. $\sec^2\theta$

c. $2\tan^2\theta$

d. 0

Difference of squares: $\sec^2\theta - \tan^2\theta = 1 + \tan^2\theta - \tan^2\theta = 1$.

© ClipSAT — clipsat.org | All Rights Reserved

35 [Intermediate] • Target: Solving Trigonometric Equations

Solve $2\cos(2x) = 1$ for $0 \le x \le \pi$.

a. $x = \frac{\pi}{6}$ **and** $x = \frac{5\pi}{6}$ ✓

b. $x = \frac{\pi}{3}$ and $x = \frac{2\pi}{3}$

c. $x = \frac{\pi}{6}$ only

d. $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$

$\cos(2x) = 1/2 \Rightarrow 2x = \pi/3, 5\pi/3 \Rightarrow x = \pi/6, 5\pi/6$.

36 [Intermediate] • Target: Solving Trigonometric Equations

Find all solutions of $3\sin x = 2\cos^2 x$ in the interval $0^\circ \le x \le 360^\circ$.

a. 30° **and** 150° ✓

b. 30° and 210°

c. 60° and 120°

d. 45° and 135°

$3\sin x = 2(1 - \sin^2 x) \Rightarrow 2\sin^2 x + 3\sin x - 2 = 0 \Rightarrow (2\sin x - 1)(\sin x + 2) = 0 \Rightarrow \sin x = 1/2 \Rightarrow x = 30^\circ, 150^\circ$.

37 [Intermediate] • Target: Solving Trigonometric Equations

Solve the equation $\tan(x + 45^\circ) = 1$ for $0^\circ \le x \le 360^\circ$.

a. $x = 0^\circ, 180^\circ, 360^\circ$ ✓

b. $x = 45^\circ, 225^\circ$

c. $x = 90^\circ, 270^\circ$

d. $x = 0^\circ$ only

$x + 45^\circ = 45^\circ, 225^\circ, 405^\circ \Rightarrow x = 0^\circ, 180^\circ, 360^\circ$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Double Angle Formulae and Graph Amplitude • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

If $\sin\theta = \frac{3}{5}$ and θ is acute, find the exact value of $\sin(2\theta)$.

a. $\frac{24}{25}$ ✓

b. $\frac{12}{25}$

c. $\frac{7}{25}$

d. $\frac{6}{5}$

$\cos\theta = 4/5$. $\sin(2\theta) = 2\sin\theta\cos\theta = 2(3/5)(4/5) = 24/25$.

39 [Challenging] • Target: Double Angle Formulae and Graph Amplitude • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

State the amplitude and period of the function $y = 3\sin(4x) - 2$.

a. **Amplitude** $= 3$**, Period** $= \frac{\pi}{2}$ ✓

b. Amplitude $= 3$, Period $= \frac{\pi}{4}$

c. Amplitude $= 1$, Period $= 2\pi$

d. Amplitude $= 4$, Period $= \frac{\pi}{2}$

Amplitude is $|a| = 3$. Period is $2\pi/b = 2\pi/4 = \pi/2$.

40 [Challenging] • Target: Double Angle Formulae and Graph Amplitude • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Express $\cos(2\theta)$ purely in terms of $\sin\theta$.

a. $1 - 2\sin^2\theta$ ✓

b. $2\sin^2\theta - 1$

c. $1 - \sin^2\theta$

d. $\sin^2\theta - 1$

$\cos(2\theta) = \cos^2\theta - \sin^2\theta = (1 - \sin^2\theta) - \sin^2\theta = 1 - 2\sin^2\theta$.

41 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A sector has radius 3 cm and angle 0.4 rad. Find the arc length.

a. 17/5

b. 3/5

c. 11/5

d. **6/5** ✓

Applying Circular Measure principles, $s = r\theta = 3 \times 0.4 = 6/5$ cm Therefore, the verified correct option is 6/5.

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Radian Measure, Arc Length and Sectors • Cross-Lesson: 6.01 Permutations and Combinations

A sector has radius 20 cm and angle 2.8 rad. Find the arc length.

a. 114/5

b. **56** ✓

c. 560

d. 28

Applying Circular Measure principles, $s = r\theta = 20 \times 2.8 = 56$ cm Therefore, the verified correct option is 56.

43 [Challenging] • Target: Trigonometric Identities and Ratios • Cross-Lesson: 7.01 Differentiation and Rates of Change

A sector has radius 30 cm and angle 1.9 rad. Find the arc length.

a. **57** ✓

b. 855

c. 58

d. 57/2

Applying Circular Measure principles, $s = r\theta = 30 \times 1.9 = 57$ cm Therefore, the verified correct option is 57.

44 [Challenging] • Target: Solving Trigonometric Equations • Cross-Lesson: 8.01 Integration and Kinematics

A sector has radius 15 cm and angle 1.2 rad. Find the arc length.

a. 9

b. 81/5

c. 19

d. **18** ✓

Applying Circular Measure principles, $s = r\theta = 15 \times 1.2 = 18$ cm Therefore, the verified correct option is 18.

45 [Challenging] • Target: Double Angle Formulae and Graph Amplitude • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

A sector has radius 7 cm and angle 3.0 rad. Find the arc length.

a. 21/2

b. 147/2

c. 10

d. **21** ✓

Applying Circular Measure principles, $s = r\theta = 7 \times 3.0 = 21$ cm Therefore, the verified correct option is 21.

© ClipSAT — clipsat.org | All Rights Reserved